BSCI201: Human Anatomy and Physiology 1
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1. Part 1 of Endocrine Study
    1.1. One unusual characteristic presented in Ellie in part 1 of the case study is her inability to focus while studying and retain information. Additionally, she is rather trembly, nervously presenting, and has difficulty sleeping. Physically, Dr. Kern describes her as being quite thin and her eyes being unusually large.
    1.2. There are many possible explanations that could be the reasoning for Ellie's difficulties. Some physiological causes may be hormonal imbalances, possible introduction of the menstrual cycle, or some type of dietary gut issues. Additionally, some psychological causes may be anxiety levels, ADHD, and or insomnia.
2. Part 2 of Endocrine Study
    2.1. The thyroid gland is in the throat just inferior to the larynx.
    2.2. The Hormones secreted by the thyroid gland are thyroxine (T4), triiodothyronine (T3), and calcitonin.
        2.2.1. Thyroxine's (T4) general actions consist of metabolism regulation, energy production, growth and development, temperature regulation, and maintenance of hair, skin, and nails.
        2.2.2. Triiodothyronine's (T3) general actions consist of metabolism regulation, energy production, growth and development, temperature regulation, cardiovascular function, and nervous system function.
        2.2.3. Calcitonin's general actions consist of calcium homeostasis, bone health, antagonist to PTH, and a role in healthy kidney function.
    2.3. Exophthalmos is related to thyroid dysfunction as it is prompted by an autoimmune issue, which promotes the inflammation of tissue cells behind the eyes. Due to the lack of space behind the eyes, the increase in volume of fatty tissue causes the protrusion bulging eye look. This is categorized as Grave's Disease, which is a type of hyperthyroidism that occurs more commonly in women than men.
    2.4. The slight swelling in Ellie's neck is a symptom of Graves' Disease, called goiter, in which the enlargement of the thyroid occurs, and shows that her hypothalamus is hyperactive.
    2.5. Ellie's thyroid gland is hyperactive because she has bulging eyes, enlargement of her thyroid (goiter) which is shown by her neck protruding, difficulty sleeping, and experiencing difficulty gaining weight due to a high metabolism rate. Due to Ellie's synonymous symptoms with Graves' disease, it is most likely that she is also exhibiting thyrotoxicosis which is excessive thyroid action.

2.6. The expected results of the blood tests if Ellie has a hyperactive thyroid there will be a lower presence of TSH hormones in the blood due to the negative feedback loop that ensues. The expected results of the blood tests, if Ellie has a hypoactive thyroid, will be elevated levels of the TSH hormones in the blood as the pituitary gland is trying to signal the thyroid to produce more thyroid hormones.

3. Part 3 of Endocrine Study
    3.1. What is causing Ellie's thyroid to secrete too much T3 and T4 is her diagnosis of Graves disease, in which her thyroid is hyperactive. This is due to the autoimmune issue in which the immune system mistakenly produces antibodies that stimulate the thyroid gland without the need for TSH.
    3.2. Ellie is incorrect in thinking that TSH is a thyroid hormone, but rather it is a thyroid-stimulating hormone, secreted by the pituitary gland, thus the presence of TSh is what prompts the promotion of true thyroid hormones like T3 and T4. Ellie's TSH levels are low instead of high as the increased levels of her T3 and T4 hormone levels are high thus the TSH hormone does not present as there is no need to increase the production of T3 and T4 because it is already present enough.
4. Part 4 of Endocrine Study
    4.1. Her diagnosis is unusual because usually people are 30-50 at the time of diagnosis, while she is only 20. That being said, the diagnosis can still affect children and older adults. However, Grave’s disease is much more common in women compared to men, thus putting her in a higher risk pool.
    4.2. Beta-blockers reduce symptoms of Grave’s disease such as increased heartbeat and anxiety by blocking beta receptors from adrenaline attachment (which causes increased heartbeat), however; they do not stop or reduce the thyroid hormone production. Beta-blockers don’t interact directly with the thyroid or hormone release, instead, it only focus on the symptoms of increased heartbeat.
    4.3. The following treatment options all have advantages and disadvantages:
        4.3.1. Anti-thyroid medications (methimazole, propylthiouracil) - medications that slow the production of thyroid hormones.
            4.3.1.1. Advantages include that it is effective in controlling hyperthyroidism, it is non-invasive, the medication works rather quickly, and is often suitable for large sectors of the population dealing with hyperactive thyroids.
            4.3.1.2. Disadvantages include risk of side effects, relapse even while on medication, regular monitoring is necessary, it is not a permanent solution, and there are pregnancy considerations.
        4.3.2. Radioactive Iodine therapy destroys the cells of the thyroid gland that produce the thyroid hormone.
            4.3.2.1. Advantages are that it can be taken by pills or by liquid medication and that it does not destroy other tissues of the body in the process.

4.3.2.2. Disadvantages are that radioactive iodine therapy, while effective, may need to be done more than once in order to bring the thyroid hormone levels back to normal. Additionally, this treatment is not used by pregnant or breastfeeding mothers since it can harm the thyroid of the fetus and it can be passed on from the mother to the baby through breastmilk.

4.3.3. Surgery - part of the thyroid gland is removed.

4.3.3.1. Advantages include the prevention of recurrence of hyperthyroid issues and reduction of the production of thyroid hormones.

4.3.3.2. Disadvantages include risk of complication, damage to nearby structures in surgery, the development of hypothyroidism, cosmetic reasons, postoperative pain, recovery time, and the of course normal surgical risk of infection.